

Geometry of Root Systems and Dynkin Diagrams

Classification of Continuous Symmetries in Physics

Theoretical Physics Master - Group Theory

Master Study Guide

Overview & Roadmap

The Goal: To demonstrate how the classification of continuous symmetries in the Universe reduces to a rigid problem of Euclidean geometry.

Roadmap:

- ▶ **Assumed Structure:** We assume the Cartan-Weyl decomposition of a semi-simple Lie algebra.
- ▶ **Lemma A (Crystallographic Restriction):** We establish the severe quantization of angles between roots.
- ▶ **Lemma B (Simple Roots):** We identify the fundamental building blocks of the algebra.
- ▶ **Main Result:** The visual classification of rank-2 algebras and the construction of **Dynkin Diagrams**.

The Root System

From the Cartan-Weyl decomposition of a semi-simple Lie algebra $\mathfrak{g}$, the roots (eigenvalues of the adjoint action) form a highly symmetric geometric structure inside a Euclidean space $E = \mathfrak{h}_{\mathbb{R}}^{\vee}$.

Physical Meaning: The roots correspond to the non-Cartan generators of our symmetry (e.g., $W^{\pm}$ bosons in the Standard Model, acting as raising/lowering operators for quantum numbers).

The rigidity of quantum mechanics (specifically $SU(2)$ representation theory) forces these roots to satisfy strict geometric axioms.

Formal Definition of Root System

The Crystallographic Restriction

The requirement that roots must fold into representations of independent $SU(2)$ subalgebras forces the quantity $n_{\alpha\beta} = \frac{2(\alpha,\beta)}{(\alpha,\alpha)}$ to be an integer.

By the definition of the Euclidean scalar product, we can relate this to the angle $\theta_{\alpha\beta}$ between two roots α and β :

$$n_{\alpha\beta} n_{\beta\alpha} = 4 \frac{(\alpha, \beta)^2}{\|\alpha\|^2 \|\beta\|^2} = 4 \cos^2 \theta_{\alpha\beta} \in \mathbb{Z} \quad (1)$$

Because $\cos^2 \theta \leq 1$, the product must be strictly bounded:

$$n_{\alpha\beta} n_{\beta\alpha} \in \{0, 1, 2, 3, 4\} \quad (2)$$

Extended Algebraic Proof

Allowed Angles and Lengths

Excluding the trivial collinear case (4), the allowed angles between any two linearly independent simple roots are heavily restricted.

For $\theta \geq \pi/2$, the only allowed geometric configurations are:

- ▶ $\theta = \pi/2$ (90°): Roots are orthogonal.
- ▶ $\theta = 2\pi/3$ (120°): Equal lengths ($\|\beta\|/\|\alpha\| = 1$).
- ▶ $\theta = 3\pi/4$ (135°): Length ratio is $\sqrt{2}$.
- ▶ $\theta = 5\pi/6$ (150°): Length ratio is $\sqrt{3}$.

Why it matters: This mathematical rigidity is the profound reason why there are only a handful of distinct continuous gauge groups (like $SU(N)$, $SO(N)$) allowed in physics!

Classification of Rank 2 Root Systems

Using the allowed angles, we can geometrically construct all possible rank 2 root systems (algebras with a 2-dimensional Cartan subalgebra).

There are exactly four mathematical solutions:

1. $A_1 \oplus A_1$ (e.g., $SU(2) \times SU(2)$)
2. A_2 (e.g., $SU(3)$, the Eightfold Way flavor symmetry)
3. $B_2 \cong C_2$ (e.g., $SO(5)$)
4. G_2 (Exceptional algebra)

[Insert Image: Rank 2 Root Systems $A_1 \oplus A_1$, A_2 , B_2 , G_2]

Simple Roots and Dynkin Diagrams

To formalize the classification for any rank, we introduce **Simple Roots** (Δ): a minimal basis of positive roots that generate the entire system.

We condense the geometry of simple roots into a **Dynkin Diagram**:

- ▶ **Nodes:** Represent simple roots.
- ▶ **Edges:** Represent the angle between them (0, 1, 2, or 3 lines).
- ▶ **Arrows:** Point from the longer root to the shorter root.

[Definition of Simple Roots](#)[Definition of Dynkin Diagrams](#)

The Main Result: Rank 2 Dynkin Diagrams

The geometric root systems directly translate into the fundamental Dynkin diagrams:

[Insert Image: Dynkin Diagrams $A_1 \oplus A_1$, A_2 , B_2 , G_2]

From these simple building blocks, one can classify all semi-simple Lie algebras, setting the stage for constructing Grand Unified Theories (GUTs).

Formal Definition: Root System

Definition 4.4.1: Let E be a finite-dimensional Euclidean vector space with inner product $(\cdot, \cdot)$. A *root system* $\mathcal{R} \subset E$ is a finite set satisfying:

1. $\text{span}_{\mathbb{R}}(\mathcal{R}) = E$ and $0 \notin \mathcal{R}$.
2. If $\alpha \in \mathcal{R}$ and $\lambda\alpha \in \mathcal{R}$ for $\lambda \in \mathbb{R}$, then $\lambda = \pm 1$.
3. **Invariance under reflection:** For all $\alpha, \beta \in \mathcal{R}$, the reflection of β along α is in $\mathcal{R}$:

$$s_{\alpha}(\beta) := \beta - \frac{2(\alpha, \beta)}{(\alpha, \alpha)}\alpha \in \mathcal{R}$$

4. **Crystallographic Condition:** For all $\alpha, \beta \in \mathcal{R}$, the projection is an integer:

$$n_{\alpha\beta} := \langle \alpha, \beta \rangle = \frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}$$

Formal Definition: Simple Roots

Definition 4.4.3: Let $t \in E$ be a vector such that $(t, \alpha) \neq 0$ for all $\alpha \in \mathcal{R}$.

- ▶ A root α is defined as **positive** ($\alpha \in \mathcal{R}^+$) if $(t, \alpha) > 0$.
- ▶ A positive root $\alpha \in \mathcal{R}^+$ is called **simple** if it cannot be written as the sum of two positive roots.

The set of simple roots is denoted by $\Delta \subset \mathcal{R}$.

Theorem 3.5.5: Every root $\alpha \in \mathcal{R}$ can be uniquely written as an integer linear combination of simple roots:

$$\alpha = \sum_{i=1}^r n_i \alpha^{(i)}$$

where all $n_i \geq 0$ (if $\alpha \in \mathcal{R}^+$) or all $n_i \leq 0$ (if $\alpha \in \mathcal{R}^-$).

Formal Definition: Dynkin Diagrams

Definition 4.4.5: A Dynkin diagram is a partially directed graph associated with a root system:

- ▶ The nodes are the simple roots $\alpha \in \Delta$.
- ▶ For $\alpha, \beta \in \Delta$ with $\|\alpha\| \leq \|\beta\|$, they are connected by $\ell_{\alpha\beta}$ edges, where:

$$\ell_{\alpha\beta} = -n_{\alpha\beta} = 4 \cos^2 \theta_{\alpha\beta} \in \{0, 1, 2, 3\}$$

- ▶ If $\ell_{\alpha\beta} > 1$ (i.e. roots have different lengths), an arrow is drawn from the longer root β to the shorter root α .

Extended Proof: Angle Quantization

We want to show that $n_{\alpha\beta}n_{\beta\alpha} \in \{0, 1, 2, 3\}$.

$$\begin{aligned} n_{\alpha\beta}n_{\beta\alpha} &= \left(\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \right) \left(\frac{2(\beta, \alpha)}{(\beta, \beta)} \right) \\ &= 4 \frac{(\alpha, \beta)^2}{\|\alpha\|^2 \|\beta\|^2} \\ &= 4 \cos^2 \theta_{\alpha\beta} \end{aligned}$$

Since $n_{\alpha\beta} \in \mathbb{Z}$ (crystallographic condition) and $\cos^2 \theta \in [0, 1]$, the product must be an integer between 0 and 4.

The case $4 \cos^2 \theta = 4 \implies \cos \theta = \pm 1$ implies the roots are collinear ($\beta = \pm \alpha$). By axiom 1 and 2, simple roots must be linearly independent, so we exclude 4 for simple roots.

Thus, $n_{\alpha\beta}n_{\beta\alpha} \in \{0, 1, 2, 3\}$.